PAIN

Accelerating discovery in pain science: the Acute to Chronic Pain Signatures program

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Abstract

Chronic pain is a complex and multifaceted disease, with biological causes distributed across tissues and bodily systems and influenced by psychological and social factors. A major aim of pain research is to find better ways to measure signals associated with chronic pain and thus predict, track, and treat pain conditions. Emerging biological measures related to the metabolic, immune, and nervous systems hold promise for identifying the physiology underlying pain and developing new treatments. Developing such measures will require multimodal datasets from large samples in combination with methodological advances. The Acute to Chronic Pain Signatures (A2CPS) Program serves as a model for such an effort, and its data will provide investigational opportunities for years to come. Acute to Chronic Pain Signatures is collecting multiomics, psychosocial, functional, and neuroimaging data from 2800 individuals before and after thoracic or knee replacement surgery. In addition to evaluating the prognostic value of previously identified biomarkers for predicting the transition to chronic pain, A2CPS will use machine learning and artificial intelligence to link these multiple data types and identify multimodal biosignatures of future pain. Open sharing of large datasets like that of the A2CPS, and models derived from them, will accelerate discovery and allow researchers to model brain, metabolic, and immune relationships relevant for multiple facets of health.

Keywords: Chronic pain, Genomics, Genetics, Metabolomics, Neuroimaging, Inflammation, Diffusion, fMRI, Lipodomics, Surgery, Postsurgical pain

Chronic pain is a major health challenge of the 21st century, affecting 1 in 5 people worldwide at every stage of life.[4] Chronic pain can arise from damage and inflammation in tissues throughout the body, and it can persist in the absence of damage or after an acute insult has resolved, becoming a debilitating condition in its own right.[9] Chronic pain alters the peripheral and central nervous systems, immune system, and metabolic and lipid signaling.[9]

Chronic pain conditions require innovative solutions to provide better treatments. Our vision for the future of pain research is to develop biomarkers—biological indicators of pain pathophysiology—that can pinpoint the drivers of pain in an individual and provide biological targets for treatment. To develop such biomarkers and new treatments based on them, chronic pain must be considered as a complex, multisystem disease.[7] Investments in pain research have provided important advances, but current biological research is often limited to small studies focused on a single data type. Creating the next generation of treatments will require a more holistic understanding and more comprehensive discovery of biomarkers across multiple systems—including the nervous, immune, autonomic, metabolic, and endocrine systems that support healthy function in every cell in the human body. Meaningful measurements of these processes are only recently available, and the most informative pain biomarkers have yet to be discovered. Discovering them will require bringing together (1) cutting-edge biological assays of thousands of relevant variables, (2) multisystem assessments in samples large enough to reliably discover and validate pain-linked variables, and (3) modern machine-learning and artificial intelligence (AI) approaches to identify patterns predictive of pain in subsets of individuals.

Ultimately, better treatments will require integration of biomarker discoveries—including neuroimaging and omics—into clinical assessments and applications. For example, blood assays may be improved by considering the cellular and tissue milieu and context-dependent protein expression. And brain imaging advances can identify brain pathways and networks with strong neurobiological relationships with pain—and could identify targets (eg, for drugs, behavioral interventions, or neurostimulation) that mediate clinical change.

The Acute to Chronic Pain Signatures (A2CPS) program serves as a model of such large-scale, multidisciplinary efforts.[8] Acute to Chronic Pain Signatures is a National Institutes of Health Common Fund program designed to provide a holistic understanding of the development and persistence of chronic pain after surgery.[2,3,7] The study follows over 2800 patients longitudinally before and after surgery (**Fig. 1**), which allows for highly powered validation of up to 40 primary biomarkers and discovery of new biomarkers in a broad array of multisystem assessments.

The A2CPS project was conceived on several foundational premises. First, measures should provide broad coverage of

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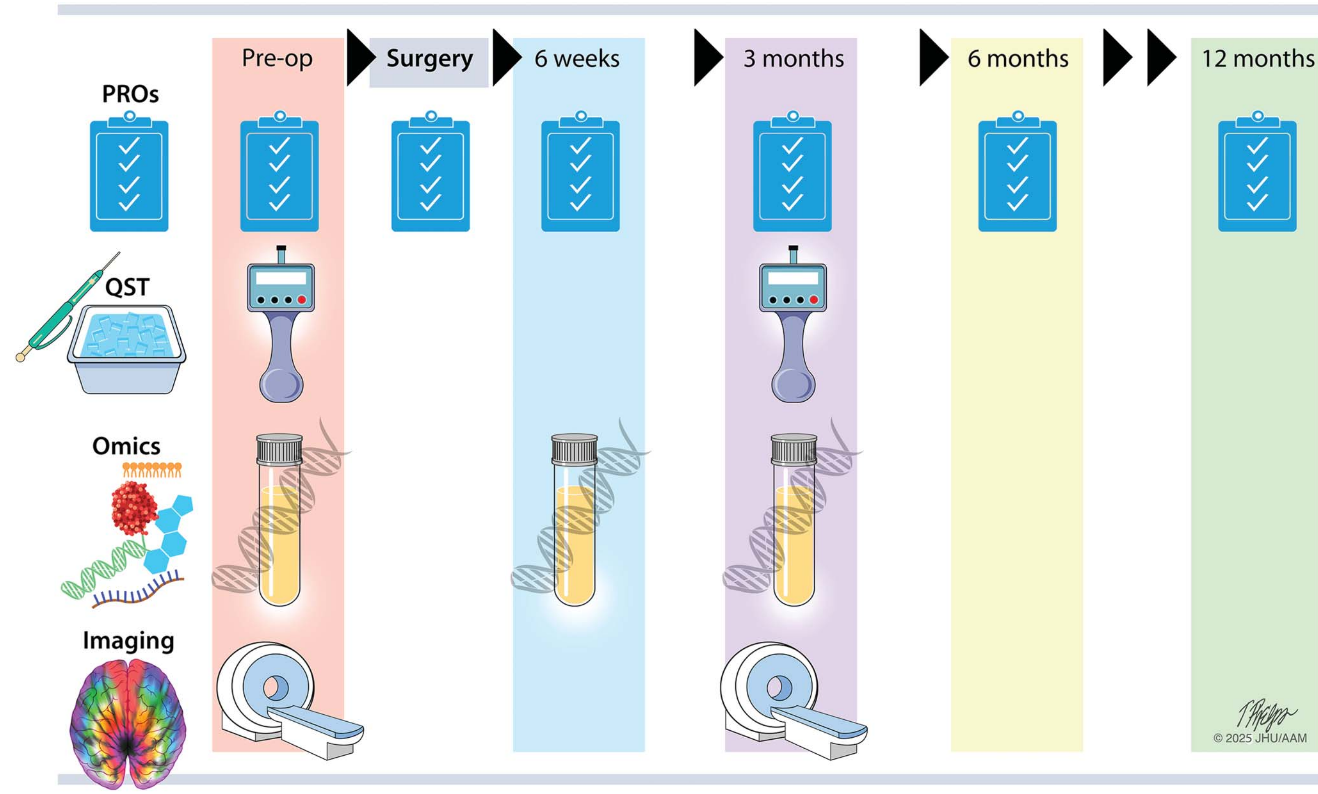

Figure 1. A2CPS protocol: all data types (PROs; QST; omics; and brain imaging) were collected from all participants before and 3 months after surgery. Omics and PROs are also collected at 6 weeks postsurgery, and PROs are collected throughout the year-long study. A2CPS, Acute to Chronic Pain Signatures; PROs, patient-reported outcomes; QST, Quantitative Sensory Testing.

potential mechanisms across brain–body systems, achieved with use of neuroimaging and multiomics panels that yield hundreds of thousands of measures from a single patient visit. Second, we must balance rigorous validation of existing candidate biomarkers with the ability to discover new biological, psychological, and behavioral predictors of chronic pain, which may be indicators of treatable biological mechanisms. This motivates the explicit statement of both discovery and validation goals, with rigorous data protections and processes established for validation. And third, we must share the data and analytical tools broadly with the scientific community, enabling exploration of diverse ideas and approaches in discovery datasets.

The A2CPS project is currently collecting and integrating data from 12 surgical centers. The measures, which will make up multimodal signatures, were selected to balance coverage of multiple systems and levels of analysis with clinical feasibility in 5 categories:

(1) Comprehensive clinical assessments of pain, function, disability, mental health, substance use, sleep, fatigue, and other health measures;
(2) Medical history from electronic health records (EHRs), including medications;
(3) Multiomics assays from blood, including genetic variants, extracellular RNA, proteomics, lipidomics, and metabolomics;
(4) Quantitative sensory testing and functional testing, including pressure pain mechanical allodynia, temporal summation, conditioned pain modulation, and performance-based function tests; and
(5) MRI neuroimaging, including an anatomical scan, diffusion-weighted scan, resting-state functional scans, and functional imaging of pressure-evoked pain.[6]

A full description of the data is available on the Web site A2CPS.org.

Biomarkers for pain can be useful in isolation and in combination. Individual biomarkers derived from a single data type could reveal an important mechanism or target. Biosignatures take a broader view, combining measures within and across modalities to optimize predictive power and effects. Ultimately, this approach has the potential to identify multiple predictors of chronic pain, which could subsequently be validated and ideally be targeted to prevent the development of chronic pain. Such approaches are used clinically for diseases including cardiovascular disease and diabetes. Using a biosignature approach, A2CPS will develop novel machine-learning algorithms to analyze the data.

The A2CPS dataset has a rich array of clinically relevant psychosocial, neuroimaging, and multiomics measures. Understanding how these data relate to one another is a critical step in building webs of meaning that make all measures more relevant and useful across research fields to address questions such as

(1) Are neuroimaging and immune correlates of pain shared with other psychological outcomes, or is pain unique?
(2) Do individuals fall into predictable subtypes, or biotypes, with different inflammatory trajectories of inflammatory responses to surgery?
(3) Are there metabolomic and lipidomic signatures of individual differences in baseline pain?
(4) Are these relationships generalizable across biological males and females?

The A2CPS dataset has sufficiently diverse measures and a large enough sample size to obtain robust answers to these questions and many others, including questions not yet conceived of.

Large public datasets such as A2CPS promise to play an important role in the next 50 years of pain research. We hypothesize that they will help drive both new methodological advances and hypothesis generation. Models trained on large, multimodal datasets can be fine-tuned to perform well on smaller, related datasets and can identify subgroups of individuals with similar pathophysiology. Integration of diverse data types will allow researchers to uncover richer patterns and boost predictive accuracy beyond what data from a single modality can achieve. Comparing and linking these patterns across data types (eg, imaging, genetics, metabolomics, and behavior) can lead to more biologically plausible and clinically meaningful insights. It also promises to allow for more individualized predictions that can aid in developing personalized treatment plans. Methodological improvements, including the use of deep learning-based AI for denoising and artifact correction, as well as improved compressed representations of complex signals in neuroimaging and -omics, will aid in these analyses. Artificial intelligence promises to be useful for identifying complex, nonlinear relationships between variables across data types. Ultimately, we believe that AI-driven multimodal analysis will help identify more nuanced subtypes of chronic pain not visible with single-modal approaches.

The A2CPS Consortium contains diverse skills and perspectives, but the curiosity and creativity of researchers across different fields will be required to realize the study's potential.[1] Acute to Chronic Pain Signatures baseline data are available to the scientific community now (https://nda.nih.gov/edit_collection.html?id=5121), and the entire, curated dataset will become available after 2027 (**Fig. 2**), ensuring that all our data are Findable, Accessible, Interoperable, and Reusable. All data are submitted to the NIMH Data Archive and all metadata to the Common Fund Data Ecosystem Data Portal, requiring well-defined data structures to describe the data and metadata. Those repositories in turn enable users to search, discover, and combine other relevant data with A2CPS information for further analysis. We use Helping to End Addiction Long-term Common Data Elements, further facilitating interoperability between data systems. In addition, we adopted the Observational Medical Outcomes Partnership common data model for our EHR data. Observational Medical Outcomes Partnership is an open, standardized data framework designed to organize and structure observational data, enhancing efficient analyses, reliable evidence production, and the ability to link EHR data from A2CPS with data from other studies. All protocols and processes used in data generation are systematically recorded to ensure the provenance of the data shared with the scientific community. These metadata are disseminated to the NIMH Data Archive and Common Fund Data Ecosystem Data Portal, enabling their use across diverse research settings. To facilitate use, the A2CPS team will release open-access online courses including instruction on how to access, understand, and analyze A2CPS data over the coming year.

Using science to understand, prevent, and alleviate chronic pain is both a moral imperative and an opportunity to advance the human

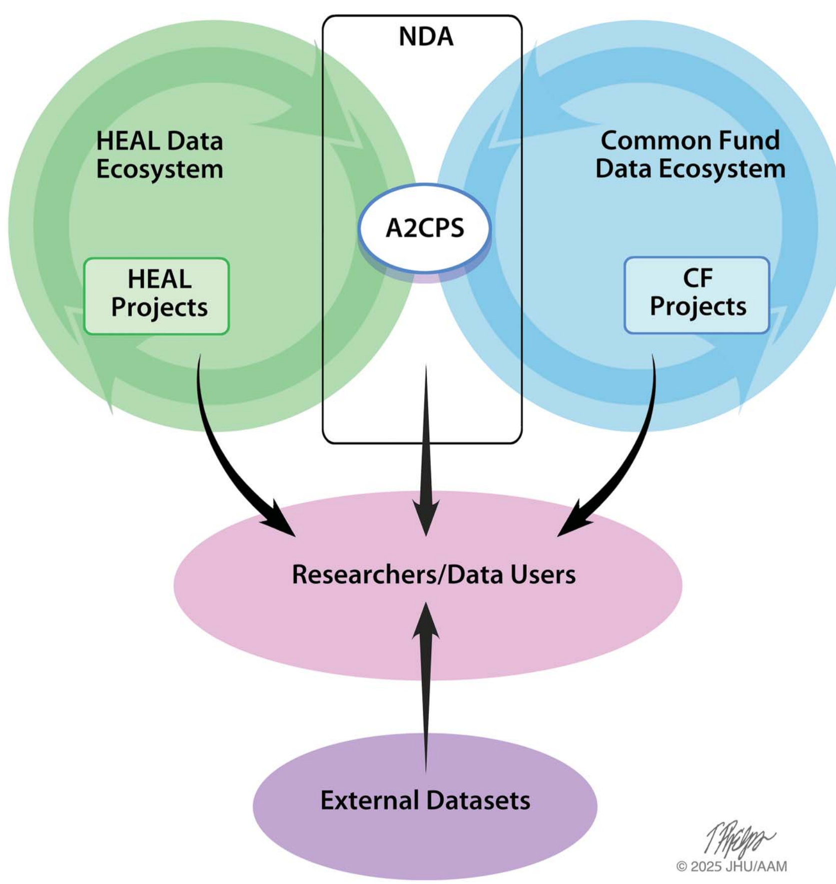

Figure 2. The open-science data-sharing ecosystem. A2CPS shares its public data releases through the NDA repository (at https://nda.nih.gov/). A2CPS metadata are also included in the NIH Common Fund data ecosystem and the HEAL data ecosystem. Harmonization provides standardized data structures and vocabularies for data elements that researchers can understand and integrate with data from other large-scale studies, including other HEAL and Common Fund studies. Users can also integrate and analyze A2CPS data with other external datasets. A2CPS, Acute to Chronic Pain Signatures; HEAL, Helping to End Addiction Long Term; NDA, NIMH Data Archive.

condition. Our vision is that one day, we can use a comprehensive assessment of pathophysiological contributors—including psychological, behavioral, nervous, immune, and metabolic—and that this assessment will guide effective, personalized treatment to prevent and manage pain. This requires bringing together data from emerging technologies to measure the human brain and body, balancing rigorous evaluation of existing methods and measures with the discovery of new biological processes. The A2CPS team of over 200 investigators and nearly 3000 participants have collectively invested hundreds of thousands of hours in this endeavor. By sharing results, models, and the entire dataset, we hope to expand this community further, welcoming your creative ideas and efforts to prevent and cure chronic pain together.

Conflict of interest statement

The authors have no conflicts of interest to declare.

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